



Official Incident Report

Event ID: 225

Rule Name: SOC257 - Vpn Connection Detected from Unauthorized Country

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Alert

The alert was triggered due to the detection of a VPN login attempt via unauthorized country belonging to the user named “Monica”. In the alert detail, VPN requests are seen via the “113[.]161.158.12” IP located in the USA.

EventID :	225
Event Time :	Feb, 13, 2024, 02:04 AM
Rule :	SOC257 - Vpn Connection Detected from Unauthorized Country
Level :	Security Analyst
Source Address :	113.161.158.12
Destination Address :	33.33.33.33
Destination Hostname :	Monica
Username :	monica@letsdefend.io
Alert Trigger Reason :	Vpn Connection Detected from Unauthorized Country
URL :	https://vpn-letsdefend.io
Show Hint 	

First, the alert should be verified by checking the available logs, and then it should be determined whether the attack was successful or not.

Detection

Verify

In Log Management, search for the attacker IP address (113[.]161.158.12) in the alert and examine the logs among the results. Firewall and Proxy logs of the attacker IP are seen on Log Management.

The screenshot shows the Log Management interface with a search bar containing the IP address 113.161.158.12. The left sidebar lists various monitoring categories, with 'Log Management' selected. The main table displays search results with columns for DATE, TYPE, SRC ADDRESS, SRC PORT, DEST. ADDRESS, DEST. PORT, and RAW. The results show a mix of Firewall and Proxy logs for the specified IP address.

DATE	TYPE	SRC ADDRESS	SRC PORT	DEST. ADDRESS	DEST. PORT	RAW
Feb, 13, 2024, 02:03 AM	Firewall	113.161.158.12	36078	33.33.33.33	443	+
Feb, 13, 2024, 02:03 AM	Proxy	113.161.158.12	36078	33.33.33.33	443	+
Feb, 13, 2024, 02:02 AM	Proxy	113.161.158.12	48075	33.33.33.33	443	+
Feb, 13, 2024, 02:02 AM	Firewall	113.161.158.12	48075	33.33.33.33	443	+
Feb, 13, 2024, 02:01 AM	Proxy	113.161.158.12	48079	33.33.33.33	443	+
Feb, 13, 2024, 02:01 AM	Firewall	113.161.158.12	48079	33.33.33.33	443	+
Feb, 13, 2024, 01:59 AM	Firewall	113.161.158.12	48990	33.33.33.33	52328	+
Feb, 13, 2024, 01:58 AM	Firewall	113.161.158.12	34703	33.33.33.33	46107	+

The VPN traffic of the attacker IP should be examined to confirm the alert. You can check the proxy logs on Log Management for the related check.

The screenshot shows the Log Management interface with a search bar containing the IP address 113.161.158.12. A modal window titled 'RAW LOG' is open, displaying the raw data for a selected log entry. The raw data shows a POST request to https://vpn-letsdefend.io/logon.html from source IP 113.161.158.12, with username Monica@letsdefend.io.

DATE	TYPE	SRC ADDRESS	SRC PORT	DEST. ADDRESS	DEST. PORT	RAW
Feb, 13, 2024, 02:02 AM	Proxy	113.161.158.12	48075	33.33.33.33	443	+
Feb, 13, 2024, 02:01 AM	Proxy	113.161.158.12	48079	33.33.33.33	443	+
Feb, 13, 2024, 02:01 AM	Firewall	113.161.158.12	48079	33.33.33.33	443	+

RAW LOG

Raw Data: Date=13/Feb/2024:02:02:13+0000, URL=https://vpn-letsdefend.io, Source IP=113.161.158.12, Request=POST, URI=logon.html, Protocol=HTTP/1.0, Response Status=200, Username=Monica@letsdefend.io

As can be seen in the log above, the request from the user "Monica[@]letsdefend[.]io" to the address "hxxps://vpn-letsdefend.io" at 02:01:01 AM received http response code 200(success).

The whois record of the relevant IP shows that it is located in Vietnam. Thus, it can be said that the alert is True Positive. However, you should check whether the attacker's attempts were successful in further analysis. In other words, receiving response code 200 to "hps://vpn-letsdefend.io" only shows that the attacker was able to access that address. You should determine whether the password entered by the attacker for the user "Monica[@]letsdefend[.]io" was correct.

Whois IP 113.161.158.12

Updated 1 second ago

```
% [whois.apnic.net]
% Whois data copyright terms    http://www.apnic.net/db/dbcopyright.html

% Information related to '113.160.0.0 - 113.191.255.255'

% Abuse contact for '113.160.0.0 - 113.191.255.255' is 'hn-changed@vnnic.vn'

inetnum:        113.160.0.0 - 113.191.255.255
netname:        VNPT-VN
descr:          Vietnam Posts and Telecommunications Group
descr:          No 57, Huynh Thuc Khang Street, Lang Ha ward, Dong Da district, Ha Noi City
country:        VN
admin-c:        PTH13-AP
tech-c:         PTH13-AP
remarks:        for admin contact mail to Nguyen Xuan Cuong NXC1-AP
remarks:        for Tech contact mail to Nguyen Hien Khanh KNH1-AP
status:         ALLOCATED PORTABLE
mnt-by:         MAINT-VN-VNNIC
mnt-lower:      MAINT-VN-VNPT
mnt-routes:     MAINT-VN-VNPT
last-modified:  2018-01-25T03:55:17Z
mnt-irt:        IRT-VNNIC-AP
source:         APNIC

irt:            IRT-VNNIC-AP
address:        Ha Noi, VietNam
```

[hxxps://www.whois.com/whois/113.161.158.12](https://www.whois.com/whois/113.161.158.12)

Analysis

IP Reputation

The attacker IP address is an external IP address. Therefore, the accuracy of the analysis can be improved by performing a reputation check. It is seen that the related IP belongs to the “Vietnam Posts and Telecommunications Group” ISP located in Vietnam.

113.161.158.12 was found in our database!

This IP was reported **1,246** times. Confidence of Abuse is **100%**: ?

100%

ISP

Vietnam Posts and Telecommunications Group

Usage Type

Unknown

Hostname(s)

static.vnpt.vn

Domain Name

vnpt.com.vn

Country

 Viet Nam

City

Hanoi, Ha Noi

IP info including ISP, Usage Type, and Location provided by IP2Location.
Updated monthly.

REPORT 113.161.158.12

WHOIS 113.161.158.12

IP Abuse Reports for 113.161.158.12:

This IP address has been reported a total of **1,246** times from 456 distinct sources. 113.161.158.12 was first reported on January 18th 2024, and the most recent report was **18 minutes ago**.



Recent Reports: We have received reports of abusive activity from this IP address within the last week. It is potentially still actively engaged in abusive activities.

Reporter	IoA Timestamp	Comment	Categories
 lukgth	2024-02-13 08:58:33 (18 minutes ago)	\$f2bV_matches	Brute-Force SSH
 pmouse	2024-02-13 08:04:12 (1 hour ago)	>20 unauthorized SSH connections	Brute-Force SSH

[hxxps://www.abuseipdb.com/check/113.161.158.12](https://www.abuseipdb.com/check/113.161.158.12)

16

/ 90

16 security vendors flagged this IP address as malicious

Similar

Graph

API

113.161.158.12 (113.160.0.0/11)

AS 45899 (VNPT Corp)

VN

Last Analysis Date

12 days ago

Community Score

DETECTION

DETAILS

RELATIONS

COMMUNITY 2

Join the VT Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Do you want to automate checks?

Antiy-AVL	Malicious	BitDefender	Phishing
Cluster25	Malicious	CRDF	Malicious
Criminal IP	Malicious	CrowdSec	Malicious
Cyble	Malicious	CyRadar	Malicious
Fortinet	Malware	G-Data	Phishing
GreenSnow	Malicious	IPsum	Malicious
Lionic	Malicious	MalwareURL	Malware
SOCRadar	Malicious	VIPRE	Phishing

<https://www.virustotal.com/gui/ip-address/113.161.158.12>

Monitoring

Log Management

Case Management

Endpoint Security

Email Security

Threat Intel

Sandbox

42806 URL

2560 IP

357 Hash

535 Domain

Select Filters...

Clear

Free text search

113.161.158.12

Date range

Select Date

Search by data type

Select

Search by data

Search

Search by tag

Search

Search

Minimize

DATE	DATA TYPE	DATA	TAG	DATA SOURCE
Dec, 07, 2023, 12:17 PM	IP	113.161.158.12	Brute Force	AbuseCH

<https://app.letsdefend.io/threath-intelligence-feed>

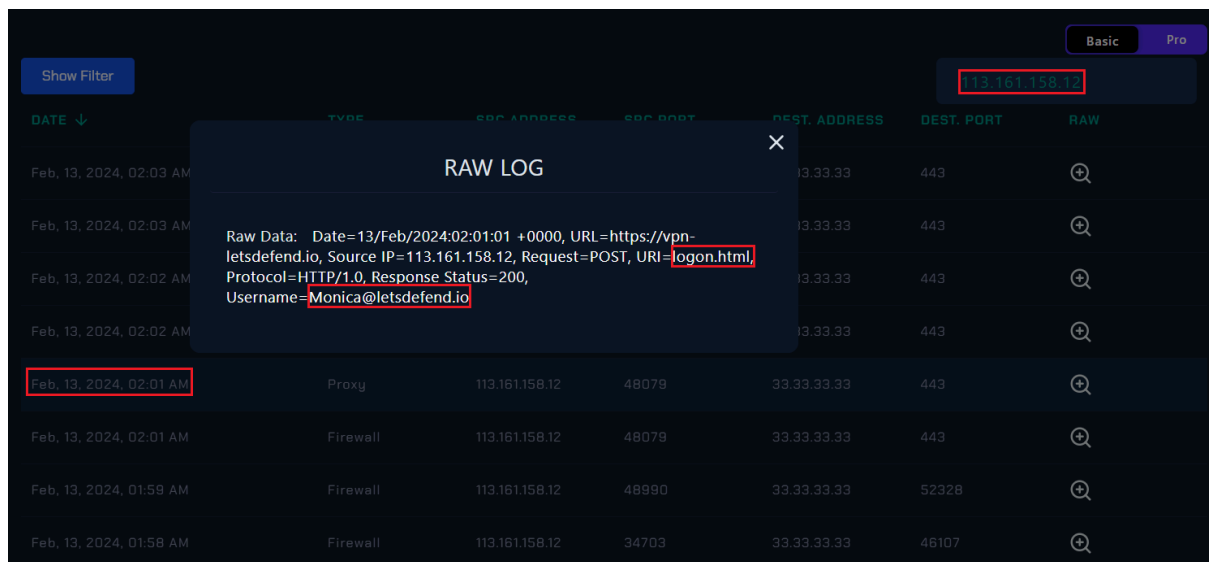
The attacker IP does not have a clean reputation according to AbuseIPDB, Virus Total, and LetsDefend TI. It has been reported by different sources at different times for various reasons. In the past, it was reported in categories such as Brute Force, Web Attack, and Malicious.

Initial Access

It was detected that the attacker attempted a VPN login via a Vietnam-based IP with the account of a user named Monica. It can be said that the External Remote Services (T1133) technique was used for initial access in this attack since the attacker attempted to access the domain via remote services such as VPN.

The issue to be investigated for this alert is whether the attacker bypassed the MFA (Multi-Factor Authentication) in the structure and, if so, how they achieved it. You should examine the logs of the attacker IP to understand this. The activities before the VPN login process should be analyzed in more detail.

It is observed in the proxy log that the attacker IP sent a request to "hxxps://vpn-letsdefend.io" at 02:01 AM and received http response code 200.



DATE	TYPE	SRC ADDRESS	SRC PORT	DEST ADDRESS	DEST PORT	RAW
Feb, 13, 2024, 02:03 AM				13.33.33	443	+
Feb, 13, 2024, 02:03 AM				13.33.33	443	+
Feb, 13, 2024, 02:02 AM				13.33.33	443	+
Feb, 13, 2024, 02:02 AM				13.33.33	443	+
Feb, 13, 2024, 02:01 AM	Proxy	113.161.158.12	48079	33.33.33.33	443	+
Feb, 13, 2024, 02:01 AM	Firewall	113.161.158.12	48079	33.33.33.33	443	+
Feb, 13, 2024, 01:59 AM	Firewall	113.161.158.12	48990	33.33.33.33	52328	+
Feb, 13, 2024, 01:58 AM	Firewall	113.161.158.12	34703	33.33.33.33	46107	+

A test check of "hxxps://vpn-letsdefend.io" shows an SSL VPN Service like the one below. This allows the attacker to attempt a VPN login with Monica.

A screenshot of a login form titled 'Login'. The form contains the text 'Please enter your username and password.' followed by three input fields: 'GROUP', 'USERNAME', and 'PASSWORD'. A 'Login' button is located at the bottom right of the form.

However, it is understood in the log after the log above that the attacker got caught in the MFA structure. The Firewall log detail at 02:01 shows "Incorrect OTP Code" as the action. It is found that the attacker entered Monica's password correctly and an OTP (One-time password) was sent. However, it is thought that the attacker could not access the medium (mail or sms) where the OTP was sent. If they could have accessed the OTP in the relevant medium, they would have successfully logged in to the VPN. Subsequent logs showed that the attacker tried again one minute later. Similarly, the OTP was entered incorrectly in both logs. Therefore the alert is True Positive for VPN connection. However, if the alert name was VPN login, it would be False Positive.

A screenshot of a firewall log interface. A 'RAW LOG' modal is open, displaying details for a log entry. The modal shows the following information: Date: 02-13-2023, 02:01 AM; Source: 113.161.158.12; Dest: vpn-letsdefend.io; User: monica@letsdefend.io; Action: Incorrect OTP Code. The background shows a table of firewall logs with columns for DATE, SOURCE ADDRESS, DEST. PORT, and RAW. The first row of the table is highlighted with a red box.

DATE	SOURCE ADDRESS	DEST. PORT	RAW
Feb, 13, 2024, 02:03 AM	113.33.33.33	443	+
Feb, 13, 2024, 02:03 AM	113.33.33.33	443	+
Feb, 13, 2024, 02:02 AM	113.33.33.33	443	+
Feb, 13, 2024, 02:02 AM	113.33.33.33	443	+
Feb, 13, 2024, 02:01 AM	113.33.33.33	443	+
Feb, 13, 2024, 02:01 AM	113.161.158.12	48079	+

You can check the emails sent to monica[.]letsdefend[.]io via Email Security. Thus, you can see whether an OTP was received. As you can see below, there are three emails with the subject "Your One-Time Passcode (OTP) for MFA Activation" containing OTP at 02:01, 02:02 and 02:03 AM.

Search Here... Q OR Detailed Search ^

Monitoring
Log Management
Case Management
Endpoint Security
Email Security
Threat Intel
Sandbox

Sender: Recipients: **monica@letsdefend.io** Subject:
Sender IP Address: Attachment Name: Email Body:
Date: 2024-01-13 to 2024-02-13 Action: Choose Action v
Search Clear Search

Result: 3 Mails

Date	Sender	Recipients	Subject	Final Action
Feb, 13, 2024, 02:03 AM	security@letsdefend.io	monica@letsdefend.io	Your One-Time Passcode (OTP) for MFA A...	Allowed
Feb, 13, 2024, 02:02 AM	security@letsdefend.io	monica@letsdefend.io	Your One-Time Passcode (OTP) for MFA A...	Allowed
Feb, 13, 2024, 02:01 AM	security@letsdefend.io	monica@letsdefend.io	Your One-Time Passcode (OTP) for MFA A...	Allowed

The details of the mail show OTPs, the IP attempting access, and various information.

From: security@letsdefend.io
To: monica@letsdefend.io
Subject: **Your One-Time Passcode (OTP) for MFA Activation**
Date: Feb, 13, 2024, 02:01 AM
Action: Allowed

Hi Monica,

To enhance the security of your account, we have initiated the Multi-Factor Authentication (MFA) activation process. As part of this process, we are sending you a One-Time Passcode (OTP).

One-Time Passcode (OTP): **18075**

Please use the provided OTP to complete the MFA activation. If you did not initiate this process or have any concerns, please contact our customer support immediately.

When and where did this happen?

Date : Feb 13, 2024 02:01:10
IP : **113.161.158.13**
URL : **https://vpn-letsdefend.io**
OS : Windows
Browser : Chrome
Location : Hanoi, Ha Noi

Best regards,
Letsdefend

Even though it is seen that the attacker failed to enter the OTP, you should check all logs belonging to the attacker. For this, search for attacker IP on Log Management. As a result, various Firewall logs were seen as shown below.

Monitoring

Log Management

Case Management

Endpoint Security

Email Security

Threat Intel

Sandbox

BasicPro

113.161.158.12

Show Filter

DATE	TYPE	SRC ADDRESS	SRC PORT	DEST. ADDRESS	DEST. PORT	RAW
Feb, 13, 2024, 01:54 AM	Firewall	113.161.158.12	33309	33.33.33.33	55997	🔍
Feb, 13, 2024, 01:56 AM	Firewall	113.161.158.12	61654	33.33.33.33	48090	🔍
Feb, 13, 2024, 01:56 AM	Firewall	113.161.158.12	57981	33.33.33.33	443	🔍
Feb, 13, 2024, 01:58 AM	Firewall	113.161.158.12	34703	33.33.33.33	48107	🔍
Feb, 13, 2024, 01:50 AM	Firewall	113.161.158.12	14683	33.33.33.33	31054	🔍
Feb, 13, 2024, 01:56 AM	Firewall	113.161.158.12	59379	33.33.33.33	11121	🔍
Feb, 13, 2024, 01:57 AM	Firewall	113.161.158.12	64441	33.33.33.33	40660	🔍
Feb, 13, 2024, 01:57 AM	Firewall	113.161.158.12	23409	33.33.33.33	48628	🔍
Feb, 13, 2024, 01:50 AM	Firewall	113.161.158.12	49905	33.33.33.33	23213	🔍
Feb, 13, 2024, 01:56 AM	Firewall	113.161.158.12	20719	33.33.33.33	40030	🔍

1234

As shown in the logs above, requests were seen from different ports. It was observed that the attacker performed port scan on IP 33[.]33.33.33.

Containment

Although the attacker was not able to bypass the MFA structure in the VPN, it can be said that Monica is a Compromised Account because the password was entered correctly. It is not necessary to isolate the system as the attacker could not gain access. However, the user's password should be reset to avoid such situations from happening again.

Lesson Learned

- It is recommended to give country or location-based access on Firewall.
- It is recommended not to use the same passwords on different platforms.
- It is recommended to set a strong password policy on clients and servers.
- It is recommended to implement a lockout policy.

Appendix

MITRE

Reconnaissance	Initial Access	Credential Access
1 techniques	1 techniques	1 techniques
Active Scanning (0/0)	External Remote Services	Multi-Factor Authentication Request Generation

Artifacts

Field	Value
Attacker IP Address	113[.]161.158.12
User/IPs	monica[@]letsdefend[.]io/172[.]16.17.163